Chapter 6

Systemic Demand Uncertainty and Chain Effects

6.1 Introduction

Supply chains are complex networks composed of entities that order and selsl products: for example a reseller or a distributor orders from a factory and resells product to other customers, that in turn might be further reselling or be final customers. In our exposition we have emphasized on the random nature of the demand. In this chapter we study quantitatively how risk propagates in a supply chain network.

Intuitively, when producing or selling a product, the optimal behavior is to produce/order the average plus a buffer to cover for the variance of the demand depending on its distribution according to the profit structure of the business in an optimal way. In a supply chain, the risk propagates and *bubble-up* throughout the network, making the risk of a stockout higher. Lead times, that is, the time orders take to be fulfilled also adds a layer of propagation of risk that could affect the supply chain adversely. We formalize these insights into a powerful paradigm (based on Stochastic Networks theory, see §7.3 [3]) that will allow us to study these effects in detail quantitatively.

6.2 Basic Model

To start, consider a single business buying/producing certain product and selling it to many different customers (potentially other businesses), all participants constitute a network with N nodes. As we studied in Chapter 2, when the demand is normally distributed (which in this case simplifies the analysis greatly), the optimal quantity for a business i is to order a quantity:

$$q_i^{NV} = \mu_i + z_i\sigma_i, \tag{6.1}$$

Where μ_i is the average demand, σ_i is the demand standard deviation and $z_i = \Phi^{-1}(\rho_i)$ is the optimal factor in the Newsvendor model, where ρ_i is the critical ratio given by the profit structure in the objective function (for example $\rho_i = (p_i - c_i)/p_i$ in the plain vanilla specification in Chapter 2.2).

Node i has an average demand μ_i that needs to be satisfied in equilibrium as:

$$\mu_i = \sum_j p_{ij} q_j + v_i, \tag{6.2}$$

Where q_j is the effective quantity produced by node j and $p_{ij} \in (0,1)$ is the proportion of the demand of j produced by node i and v_i is the external average demand. Then, assuming independence between the nodes (here the variance is assumed exogenous from the external demand), the variance σ_i^2 is equal to:

$$\sigma_i^2 = \sum_j p_{ij}^2 \sigma_j^2 + w_i^2, \tag{6.3}$$

Where w_i^2 is the *external variance* of the demand (which can be 0 if node i does not serve an external demand). Then, the effective quantity that node i needs to produce is given by:

$$q_i = \min\left(\sum_j p_{ij} q_j + \alpha_i, C_i\right), \tag{6.4}$$

Where $\alpha_i = v_i + z_i\sigma_i$. When the constraint holds with equality, this is the optimal quantity according to the Newsvendor model (as $q_i = \mu_i + z_i\sigma_i = \sum_j p_{ij} q_j + \alpha_i$) or the maximum physical capacity C_i of node i. Since this is true for all nodes in the network, then, the maximum flow of product in the supply chain network is obtained by solving the following LP:

$$\begin{aligned} \max \quad & |q| \\ \text{s.t.} \quad & (I - P)q^\intercal \leq \alpha^\intercal, \\ & \mathbf{0} \leq q \leq C. \end{aligned} \tag{6.5}$$

Where P is a sub-stochastic matrix with entries p_{ij} and α is a vector with entry i equal to $\alpha_i = v_i + z_i\sigma_i$. For most nodes v_i and w_i will be assumed to be 0 (think of producers/resellers that do not sell directly to consumers) and it will be positive for retailers (or nodes that face external demand directly). C is a vector with entry i equal to C_i, the capacity of that node. In Figure 6.1 we see an example of a supply chain network with a producer, resellers and retailers.

To solve the problem there are multiple strategies: the simplest one is to solve directly the LP obtaining a solution q^* for the maximum flow of product. Since there are N nodes, a basic feasible solution of the problem will have exactly N binding constraints in the optimal solution

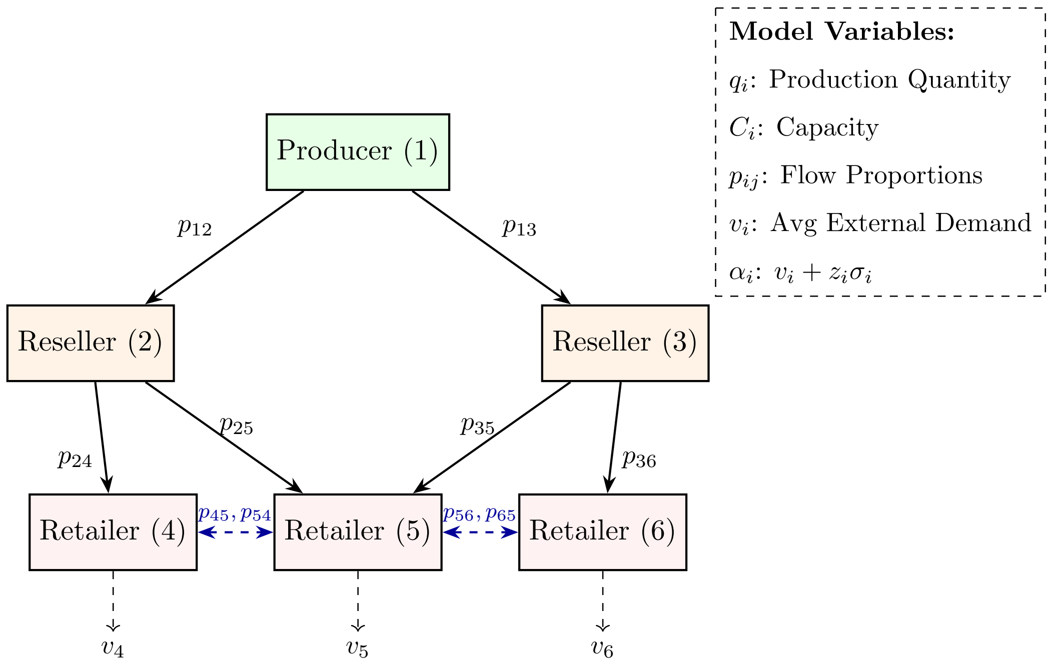

Figure 6.1: Example of a supply chain network with a single producer (represented by a node 1), 2 resellers (nodes 2,3) and 3 retailers (nodes 4,5,6) that potentially sell to each other as well as servicing the external demands (v_4, v_5, v_6).

(ignoring potential degeneracies). For example, for the constraints $q \leq C$ the nodes will be divided into two sets: one set of nodes $U \subseteq \{1, \ldots, N\}$ that will be producing at capacity $q_U = C_U$, these nodes are *un-hedged* against the risk implied by the Newsvendor model. In other words, they are limited by their capacity and with a non-negligible chance might suffer a stock-out of product. Conversely, the other set of nodes $\mathcal{H} = \{1, \cdots, N\} \setminus U$ is producing at the maximum possible given the external demand and the structure of the network.

Then, the other flow constraints should hold with equality for the other variables, that is, $(I_{\mathcal{H}} - P_{\mathcal{H}})q_{\mathcal{H}}^\intercal + (I_{\mathcal{H},U} - P_{\mathcal{H},U})C_U^\intercal = \alpha_{\mathcal{H}}^\intercal$. Which simplifies to $q_{\mathcal{H}}^\intercal = (I_{\mathcal{H}} - P_{\mathcal{H}})^{-1}(\alpha_{\mathcal{H}}^\intercal + P_{\mathcal{H},U}C_U^\intercal)$. Then, it must be that the optimal is given by:

$$q_U^\intercal = C_U^\intercal, \quad q_{\mathcal{H}}^\intercal = (I_{\mathcal{H}} - P_{\mathcal{H}})^{-1}(\alpha_{\mathcal{H}}^\intercal + P_{\mathcal{H},U}C_U^\intercal). \tag{6.6}$$

The optimality of this solution can be verified using duality checking that the solution satisfies the complimentary slackness conditions.

More importantly, the solution highlights the structure of the problem: namely, some nodes will be capped by their capacity and will not be producing enough to optimally hedge their risk.

6.2.1 The bullwhip effect and other sensitivities

In the supply chain literature there is an empirical principle known as the *bullwhip effect*. The bullwhip effect is the downstream increase in demand throughout the supply chain network caused by an increase in customer demand, which in turn ripples increasingly throughout the network.

In our model this and other effects can be quantified, we will see that the bullwhip effect is not only caused by increases in demand, but it is also exacerbated by the structure of the supply chain network and how it also propagates the variance of the external demands. The first step is to calculate the sensitivities, which is a fancy name for the derivatives of the solution with respect to some variables. The sensitivity with respect to α_i (capturing both the sensitivity to the demand and the volatility) is given by:

$$\frac{\partial q_{\mathcal{H}}}{\partial \alpha_i} = (I_{\mathcal{H}} - P_{\mathcal{H}})^{-1} e_i^{\intercal} \quad \text{for } i \in \mathcal{H}, \tag{6.7}$$

$$\frac{\partial q_{\mathcal{H}}}{\partial C_j} = (I_{\mathcal{H}} - P_{\mathcal{H}})^{-1} P_{\mathcal{H},U} e_j^{\intercal} \quad \text{for } j \in U. \tag{6.8}$$

Where e_i is a basis vector with entry i equal to 1 and the rest equal to zero. In both sensitivities, we can see the role of the matrix $(I_{\mathcal{H}} - P_{\mathcal{H}})^{-1}$. This matrix is commonly called the *network multiplier* matrix. Intuitively, recall that for a geometric series we have the well known result:

$$1 + \delta + \delta^2 + \delta^3 + \cdots = (1 - \delta)^{-1}, \tag{6.9}$$

Which is true as long as $|\delta| < 1$, which intuitively means that the terms δ^k for $k \to \infty$ not only decrease to 0, but their order of magnitude vanish as the partial sum goes to infinity (the mathematical notion of convergence of a series). Imagine the following situation to get some intuition: An oil producer, generages a quantity $q = D + 5\%q$ of oil, that is, their external demand D plus 5% of their own output that they consume to re-refine the oil. Rearranging the terms yields $q = \frac{D}{1-5\%} = D(95\%)^{-1}$. The total quantity is the demand multiplied by a *feedback* factor $(95\%)^{-1} \approx 105.3\%$. This can be seen as a feedback loop in the production caused by output that reenters the production process. The same happens in a network case with matrices, where we have that:

$$I + P + P^2 + P^3 + \cdots = (I - P)^{-1}, \tag{6.10}$$

It is also expected that for the result to hold, the higher powers of P^k go to 0 as $k \to \infty$ and vanish from the partial sums. Luckily, this holds true for substochastic matrices in our base model[1]. Moreover, the same intuition applies, each product of the matrix can be seen as a feedback loop of an increase of the required production coming from neighboring nodes, that in turn increase the production requirements for their neighbor nodes. This back and forth stabilizes in the form of multipliers (like in the above paragraph example of $(95\%)^{-1}$), and the entries of $(I_{\mathcal{H}} - P_{\mathcal{H}})^{-1}$ represent exactly the multiplier of node in a row to a node in a column. That is why the sensitivity $(I_{\mathcal{H}} - P_{\mathcal{H}})^{-1} e_i^{\intercal}$ is the row sum of of all columns entering

[1] In general, what is required for the result to hold is that the largest eigenvalue of the matrix is less than 1 in absolute value.

into node $i \in \mathcal{H}$. Likewise, the sensitivity $(I_{\mathcal{H}} - P_{\mathcal{H}})^{-1} P_{\mathcal{H},U} e_j^\intercal$ represents the weighted (by $P_{\mathcal{H},U}$) multipliers of the hedged nodes into the unhedged ones.

By the chain rule, it is straightforward to calculate the effect of a change in the demand as $\frac{\partial q_{\mathcal{H}}}{\partial v_i} = \frac{\partial q_{\mathcal{H}}}{\partial \alpha_i}\frac{\partial \alpha_i}{\partial v_i}$ and $\frac{\partial \alpha_i}{\partial v_i} = 1$ recalling $\alpha_i = v_i + \sigma_i z_i$. Therefore, $\frac{\partial q_{\mathcal{H}}}{\partial v_i} = (I_{\mathcal{H}} - P_{\mathcal{H}})^{-1} e_i^\intercal$ is the same sensitivity as before.

The bullwhip effect is known as the empirical phenomenon when an increase of the demand ripples with increasing magnitude accross the supply chain. With our framework, we can study permanent increases in the demand, as well as increases in the volatility that as we will see, quantitatively ripple differently in the network. Suppose, the demand in one of the customer facing nodes increases, that is, v_i increases for some node i. We are then interested in quantifying the impact of this increase in the demand.

To make the analysis easier, suppose the structure of the network is as follows: the graph is an acyclic directed graph and the matrix P is substochastic, where nodes are ordered in a way that they provide their output to nodes numbered higher than them. For example, in the graph in Figure 6.1 the node 1 represents a producer, which supplies their output to the reseller nodes 2 and 3. In turn, node 2 sells their output to retailers 4 and 5, and node 3 sells to 5 and 6. Note that the retailer nodes, 4 sells to 5, and 5 sells to 4 which violates our assumption. So, imagine the same network, but without node 5 selling to 4 and withouth 6 selling to 5. Later we will see that this is not really a necessary condition, but it makes the next argument easier to explain.

Imagine there is a permanent shock to the demand in a node that increases it to a new level, for example, in the graph in Figure 6.1 without the cycles, imagine that node 6 demand increases to a new level $v_6' = v_6 + \epsilon$ from the previous v_6. This increase affects first node 6, which increases their production by an amount ϵ. Next, the nodes that are affected are nodes 3 (a reseller) and 5 (another retailer). These in turn increase their demand by a magnitude ϵp_{36} and ϵp_{56} respectively. This continues happening so forth throughout all the network (and note that sometimes the effect will go forward again, as after node 3 increases their production, so will the output that node 5 does).

A way to analyze this effect numerically, is to look at the *network multipliers.* Moreover, suppose that all nodes have infinite capacity $C_i \to \infty$, so that all nodes are producing at their hedged level, so that $q = (I - P)^{-1}\alpha^\intercal$. Since the network is free of cycles, the multiplier matrix $(I - P)^{-1}$. The multiplier effect of a shock in node j into node i is the element $[(I - P)^{-1}]_{i,j}$. In the expanded form this is:

$$[(I - P)^{-1}]_{i,j} = I_{i,j} + P_{i,j} + P_{i,j}^2 + \cdots + P_{i,j}^{N-1}, \tag{6.11}$$

First note that there are N elements in the summation, as there no longer than N steps between 2 nodes. Next, see that each term exactly reflects how an increase in the demand travels throughout the network. The first node affected is $I_{i,j}$ that will feel the full effect only if $i = j$ (for example, the node 6 in the previous paragraph). Next, the effect is the nodes that can be reached by one step $P_{i,j}$ (the nodes 3 and 6). Next, the ones that can be reached exactly in two steps $P^2_{i,j}$, in 3 steps $P^3_{i,j}$ and so on (in fact, the same analysis holds for matrices with cycles, instead of $N+1$ there is a large enough term K from which further terms vanish).

This is exactly what the bullwhip effect is, nodes that are directly affected see a lower order of magnitude effect, than deeper nodes (producers or reseller) as the shock is transmitted and amplified throughout all the network.

6.2.2 Volatility Propagation

In the literature the Bullwhip effect is also associated to the propagation of variance throughout the supply chain (affecting again the innermost nodes rather than the consumer facing ones). In our model this mechanism is captured by the structure of the variance. We can write the variance Equation in (6.3) as:

$$(\sigma^2)^\intercal = (P \odot P)(\sigma^2)^\intercal + (w^2)^\intercal, \tag{6.12}$$

Where $\sigma^2 = (\sigma_1^2, \ldots, \sigma_N^2)$, $w^2 = (w_1^2, \ldots, w_N^2)$ and $\odot$ is the element-wise product of two matrices of the same dimension, such that the i,j-th element of the matrix $(P \odot P)$ is p_{ij}^2. The equation becomes $(I - (P \odot P))(\sigma^2)^\intercal = (w^2)^\intercal$, or:

$$(\sigma^2)^\intercal = (I - (P \odot P))^{-1}(w^2)^\intercal. \tag{6.13}$$

This expression should be already familiar from the previous exposition. The matrix $(I - (P \odot P))^{-1}$ is again a multiplier, but this time of the variance, denote as $M_\sigma = (I - (P \odot P))^{-1}$. Similar to the previous case, this shows how the variance of the external demand w^2 propagates inside the network. Taking element-wise square root in both sides yields the equivalent expression $\sigma^\intercal = \sqrt{M_\sigma (w^2)^\intercal}$.

The change in the variance of nodes by a change in the external demand standard deviation w_j in node j is given by:

$$\frac{\partial \sigma^\intercal}{\partial w_j} = \operatorname{diag}\left(1/\sqrt{M_\sigma (w^2)^\intercal}\right) m_j w_j, \tag{6.14}$$

Where m_j is a column vector of M_σ. Then, we can finally express the sensitivity of the demand with respect to changes in the external variance. We have by the chain rule that $\frac{\partial q_{\mathcal{H}}}{\partial w_i} = \frac{\partial q_{\mathcal{H}}}{\partial \alpha_i}\frac{\partial \alpha_i}{\partial w_i}$ and in turn $\frac{\partial \alpha_i}{\partial w_i} = z_i \frac{\partial \sigma_i}{\partial w_i} = z_i \frac{m_{ii}}{\sqrt{M_\sigma (w^2)^\intercal}_i} w_i$ or:

$$\frac{\partial q_{\mathcal{H}}^\intercal}{\partial w_i} = (I_{\mathcal{H}} - P_{\mathcal{H}})^{-1} e_i^\intercal \left[\frac{m_{ii}}{\sqrt{M_\sigma (w^2)^\intercal}_i} z_i w_i\right]. \tag{6.15}$$

The intuition is quite interesting: on one hand there is a network multiplier effect as before, but there is a further variance multiplier effect (in square brackets) that further exacerbates a shock to the standard deviation of the external demand. Note that the shock is also proportional to the current standard deviation w_i, accentuating shocks even further.

Outdated interpretations of the Bullwhip effect attribute it to the lack of information and coordination between nodes or noise coming from forecasting and lead times. Suggesting that the lack of better information systems is the main reason behind the bubbling-up of demand shocks.

In this chapter we showed mathematically that even with perfect information and coordination (all the nodes in the network can see the production quantities and uncertainty of adjacent nodes), **the Bullwhip effect is an unavoidable and an inherent topological property of supply chain networks**. Even in the perfect information case, with instantaneous order fulfillment, hedging against risk implies the Bullwhip effect.

Example 6.1 (Rare earth supply chain). A supply chain network composed of a producer (node 1) and a seller/country (node 2). The external average demand $v_2 = 100$ tons per year, the external standard deviation of the demand $w_2 = 10$ tons. The seller buys all of its product from the producer, that is $p_{12} = 1$, the routing matrix is:

$$P = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

With a safety level $z_1 = z_2 = 1.645$ what are the optimal production quantities in this network? While maintaining the same average demand, how much an increase of 1 ton of the standard deviation of the demand increases the quantities for the producer?

For the first question, we have the following quantities:

$$\begin{aligned} q_2 = v_2 + z_2 w_2 &= 100 + 1.645(10) = 116.45 \\ q_1 = p_{12} q_2 + z_1 \sqrt{p_{12}\sigma_2^2} &= 116.45 + 1.645(10) = 132.9 \end{aligned}$$

Even when the network only has 2 nodes, the producer has to produce way above node 1 to be optimally hedged against the demand fluctuation of the seller.

For the second question, we make use of the network multiplier matrix:

$$(I-P) = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \Rightarrow (I-P)^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

Also note that $P = P \odot P$, then the network multiplier matrix $(I-P)^{-1}$ and the variance multiplier M_σ are the same. For node 1 we have that the sensitivity is:

$$\frac{\partial q_1}{\partial w_2} = [(I-P)^{-1}]_{1,2} \times \left[\frac{m_{22}}{\sigma_2} z_2 w_2\right] = 1 \times \left[\frac{1}{10} 1.645(10)\right] = 1.645 \tag{6.16}$$

Then, an increase of the demand volatility by 1 ton increases production by 1.645 tons in the producer node.

6.3 Dynamic Model

In practice, knowing only the new equilibrium quantities is important, but also the timing on how the network adapts to shocks and their magnitude. In this section we extend our model to a dynamic setting to answer questions related to the timing and magnitude of the *Bullwhip effect* and other shocks. The standard way to incorporate a dynamic version, is to solve a so-called *Skorokhod Problem* (see [7]).

Consider the evolution of the inventory if there are no capacity constraints, a so-called *free inventory process*:

$$X_i(t) = Z_i(0) + (\text{Production}_i - \text{Internal Demand}_i - \text{External Demand}_i)t, \tag{6.17}$$

Where $Z_i(0)$ is an initial level of inventory (not to be confused by the safety margin z_i). Call the expression in parenthesis θ_i showing the net-rate that the inventory changes. In matrix-form, the free inventory process can be written as:

$$X(t) = Z(0) + \theta t, \tag{6.18}$$

In this case, $\theta = C - \alpha - (\lambda \wedge C)$ as the maximum production capacity of a node is their capacity, that is, $\lambda \leq C$, recalling $C = (C_1, \ldots, C_N)$. θ_i denotes the rate the inventory increases if $\theta_i > 0$ or decreases $\theta_j < 0$, eventually reaching 0. Let λ be the effective production rate of each node.

Let $Z(t)$ be the actual inventory process. As the inventory cannot be negative, that is, $Z(t) \geq 0$, the Skorokhod problem trick is to create a function $Y(t)$ that cancels out when the inventory is negative. As this must happen for all the nodes we have the relation:

$$Z(t) = X(t) + Y(t) - Y(t)P = X(t) + Y(t)(I - P), \tag{6.19}$$

In this case, $Y(t)$ appears first as it is cancelling out when $X(t)$ is negative, but it also must account when other nodes have a shortfall that affects node i, therefore the need for the term $-Y(t)P$, or $-\sum_j Y_j(t)p_{ji}$. This keeps track of the propagation when other nodes have shortfalls, which is a mathematical way of tracking the contagion between nodes.

It must also be the case that when $Z(t) > 0$, then $dY(t) = 0$, because there is still enough inventory to satisfy incoming orders. Likewise, when $Z(t) = 0$, then $dY(t) > 0$, meaning that there is not enough production capacity to satisfy the demand. All the conditions can be summarized in the program:

$$\begin{aligned} & Z(t) = X(t) + Y(t)(I - P), \qquad (6.20) \\ \text{s.t.} \quad & Y(0) = 0, \;\; \dot{Y}(t) \geq 0, \;\; Z(t) \geq 0, \;\; Z(t)\dot{Y}(t) = 0. \end{aligned}$$

Where $\dot{Y}$ denotes the time derivative of Y, that is, $\dot{Y}(t) = dY(t)/dt$. In the long term, it must be the case that the equilibrium of the dynamic model is the same as the static one. To see why, first considering the case where all the production capacities are large enough so that any production level is possible. In this case, the evolution of the system is given by the equation:

$$\dot{Z}(t) = \dot{X}(t) + \dot{Y}(t) = \theta + \dot{Y}(t)(I - P), \tag{6.21}$$

If $Z(t) > 0$ ends in a positive level, then $dY(t) = 0$. Likewise, if $Z(t)$ reaches an equilibrium point, then $dZ(t) = 0$. Leading to the equation:

$$0 = \theta = C(I - P) - \alpha \underset{C=\lambda}{\Rightarrow} \lambda = \alpha(I - P)^{-1}, \tag{6.22}$$

Where λ represents the effective production rate of each node. In the limit, this is the same production quantity that we derived in the static case. That is $\lim_{t\to\infty} \lambda(t) = q^*$. In practice, we will see that because of their physical capacity, some nodes will be unable to produce their required quantity and they will be again partition into disjoint hedged $\mathcal{H}$ and un-hedged U nodes. In the case where nodes hit a capacity limit, this refines to the so-called *flow equation*:

$$\lambda = \alpha + (\lambda \wedge C)P, \tag{6.23}$$

Where $\wedge$ is the element-wise minimum operator. When $\lambda \wedge C = \lambda$ the previous equation holds.

Following our previous analysis, at any point t the nodes are divided into two disjoint sets of hedged $\mathcal{H}$ and un-hedged U sets. For a given time t that nodes are divided into hedge $\mathcal{H}$ and un-hedged sets U. Then, the flow Equation 6.23 refines to:

$$\lambda_{\mathcal{H}} = \alpha_{\mathcal{H}} + \lambda_{\mathcal{H}} P_{\mathcal{H}} + C_U P_{U,\mathcal{H}}, \tag{6.24}$$

$$\lambda_U = \alpha_U + \lambda_{\mathcal{H}} P_{\mathcal{H},U} + C_U P_U, \tag{6.25}$$

Solving yields the following flows at time t:

$$\lambda_{\mathcal{H}} = (\alpha_{\mathcal{H}} + C_U P_{U,\mathcal{H}})(I - P_{\mathcal{H}})^{-1}, \tag{6.26}$$

$$\lambda_U = \alpha_U + (\alpha_{\mathcal{H}} + C_U P_{U,\mathcal{H}})(I - P_{\mathcal{H}})^{-1} P_{\mathcal{H},U} + C_U P_U, \tag{6.27}$$

Note that as time moves forward, the set of hedged nodes might decrease as a result of increased demand. The rates the inventory is accumulated are given by:

$$\theta = C - \alpha - (\lambda \wedge C)P = C - \lambda \tag{6.28}$$

6.3.1 Sequential dynamics

In principle, we have all the ingredients to describe the evolution over time of the supply chain network. The nodes start at $t = 0$ with some inventory position $Z(0) \geq 0$ with production rates $\lambda(t)$ dividing the nodes into disjoint sets $\mathcal{H}(t)$ and $U(t)$. Suppose, the sets are initially known as well, that is $\mathcal{H}(0)$ and $U(0)$. Then, we just need to describe the evolution of the system until the time τ when the sets change (possibly because a set of nodes had to increase production until a time that their production limit is reached). When that time τ is reached we can just restart the clock and simulate the system again.

The evolution of the system is linear between time 0 and the time where the sets of nodes change. For all nodes already at capacity $Z_U(0) = 0$ and $Z_{\mathcal{H}}(0) > 0$. We also have the following dynamics for the hedged nodes:

$$Z_{\mathcal{H}}(t) = Z_{\mathcal{H}}(0) + \theta_{\mathcal{H}} t + Y_U(t)(I - P_{U,\mathcal{H}}) \tag{6.29}$$

Where $Y_U(t) = (\lambda_U - C_U)t$ and $\theta_{\mathcal{H}} = C_{\mathcal{H}} - \lambda_{\mathcal{H}}$.

The system keeps evolving linearly according to these dynamics until the time τ where a node in the set $\mathcal{H}$ cannot satisfy all their production obligations. This time is given by:

$$\tau = \min_{j \in \mathcal{H}, \theta_j < 0} \left\{ \frac{Z_j(0)}{-\theta_j} \right\}, \tag{6.30}$$

This follows from the fact that a node in $\mathcal{H}$ with $\theta_j < 0$ implies that $C_j < \lambda_j$, meaning that even at maximum capacity they will eventually reach their production limit. This will happen when $Z_j(0) + \theta_j t = 0$ or at $t = -Z_j(0)/\theta_j$.